

# Michael Samaan Anwar

## Data Scientist

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### Summary

Computer Science student and Machine Learning Engineer specializing in Artificial Intelligence, Computer Vision, and Data Engineering. Engineered and deployed end-to-end ML models, real-time computer vision systems, and cloud-integrated REST APIs using Python, PyTorch, TensorFlow, and modern MLOps practices. Proven track record in problem-solving, feature engineering, and data pipeline optimization.

### Education

Modern Academy, Maadi

Graduate  
Cairo, Egypt

### Experience

**Machine Learning Engineer Trainee (Microsoft Track), DEPI – Digital Egypt Pioneers Initiative** 07/2026 - Present

- Training on the Microsoft Machine Learning Engineer track of Egypt's national Digital Egypt Pioneers Initiative, an applied programme focused on building production-oriented machine learning skills

**AI Training Program Participant, Sharks** 07/2023 - 09/2023

- Explored foundational machine learning concepts and supervised classification models, implementing algorithms such as Support Vector Machines (SVM).
- Formulated data exploration and visualization pipelines, completing a capstone classification project on the Iris benchmark dataset to predict species characteristics.

**AI Training Program Participant, IT Gate** 07/2022 - 09/2022

- Completed an intensive 2-Months foundational training on core Python development and interactive environment setups using Jupyter Notebooks.
- Practiced foundational data analysis techniques using NumPy and Pandas for data manipulation, structured cleaning, and introductory statistical visualization.

### Projects

#### Football Computer Vision System

Technologies: Python, OpenCV, YOLOv8, ByteTrack, PyTorch

- **Trained** a YOLOv8s object detection model across 3,052 images and 4,835 instances, achieving 84.8% mAP50 and 84.9% Precision.
- **Implemented** ByteTrack multi-object tracking to process real-time match analytics at 25 FPS.

#### Football Scout REST API

Technologies: Python, Flask, XGBoost, Scikit-Learn, Docker

- **Built** an ML-driven scouting platform analyzing 1,469 professional players, achieving 99.2% recommendation accuracy.
- **Deployed** an XGBoost valuation model ( $R^2 = 0.753$ ,  $MAE = \text{€}6.29M$ ) via a Flask REST API interface.

#### Expected Goals (xG) Shot Quality Pipeline

Technologies: Python, Scikit-Learn, Pandas

**Designed and evaluated** classification models estimating football shot quality and probability

#### NYC Airbnb Market Data Analysis

Technologies: Python, Pandas, NumPy, Power BI, Excel

- Cleaned and structured 40,000+ raw market listings, engineering custom location and pricing feature sets to analyze urban rental trends.
- Built interactive Power BI dashboards with slicers, KPI cards, and custom DAX measures to surface spatial pricing patterns and dynamic revenue drivers.

### Skills

**Programming:** Python, C++, SQL, HTML/CSS

**Machine Learning and AI:** Scikit-Learn, PyTorch, TensorFlow, XGBoost, Random Forest, Feature Engineering

**Computer Vision & Data Engineering:** OpenCV, YOLO (YOLOv8), Object Detection, Multi-Object Tracking (ByteTrack)

**Data and Analytics:** Pandas, NumPy, Matplotlib, Seaborn, Power BI (Power Query, DAX)

**Databases and Backend:** MySQL, REST APIs, Flask

**Cloud and Tools:** GitHub, VS Code

**Soft Skills:** Technical Problem-Solving, Analytical Thinking, Data-Driven Strategy, Agile Collaboration, Adaptability

## **Certifications**

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- Supervised Machine Learning: Regression and Classification – Stanford / DeepLearning.AI [[Credential](#)]
- Python for Data Science & AI – IBM [[Credential](#)]
- Databases and SQL for Data Science with Python – IBM [[Credential](#)]
- Power BI Business Analytics – UC Davis

## **Languages**

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**Arabic** native ·

**English**